

3.02 MWp Utility-Scale Solar PV Power Plant

Comprehensive Stakeholder DPR Addendum & Detailed Engineering Architecture

Location: Santuri, Purulia District, West Bengal | Off-taker: WBSEDCL | Interconnection: 33 kV Dedicated Feeder

DC INSTALLED
ARRAY

3.02 MWp

AC COMPACT
STATION

**4.3 MVA
Skid**

TOTAL CAPEX
INVESTMENT

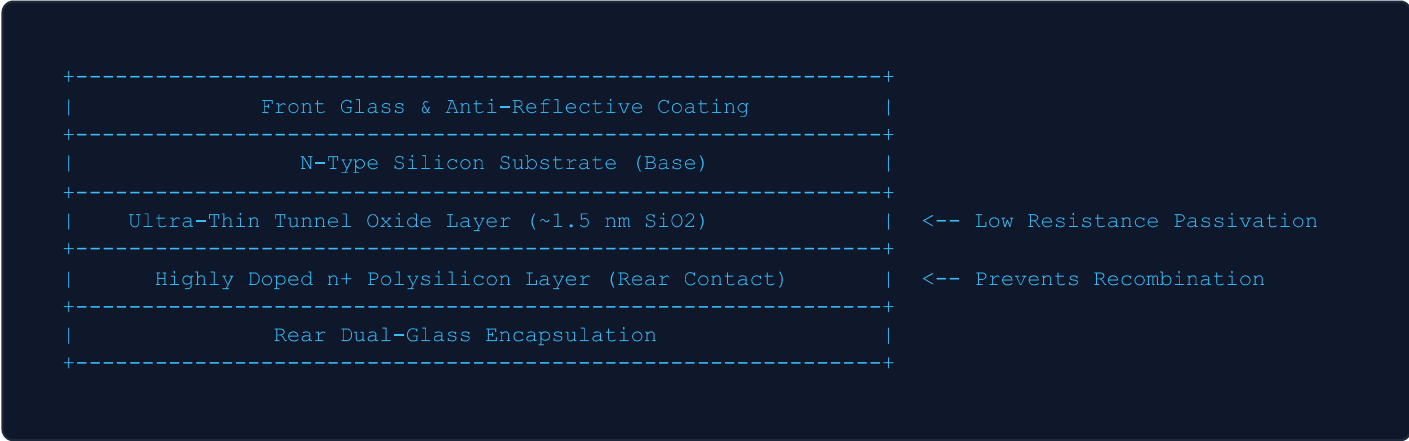
₹10.80 Cr

EQUITY / DEBT
SPLIT

30% / 70%

1.1 Photovoltaic Cell Physics & Semiconductor Dynamics

The plant utilizes **Waaree Energies 700W N-Type TOPCon (Tunnel Oxide Passivated Contact) Bifacial Modules**. N-type TOPCon technology replaces traditional P-type PERC architectures to eliminate Light-Induced Degradation (LID) and Light and elevated Temperature Induced Degradation (LeTID).



- **Passivation Physics:** TOPCon integrates an ultra-thin tunnel oxide layer (~1.5 nm) combined with a thin film of heavily doped n+ polysilicon on the rear surface. This tunnel oxide allows majority carriers (electrons) to tunnel through while passivating surface defects, reducing carrier recombination velocity to < 10 cm/s.
- **Temperature Coefficient:** The temperature coefficient of maximum power (γ) is -0.30%/°C, outperforming PERC (-0.35%/°C). This preserves generation efficiency under Santuri’s peak summer ambient temperatures of up to 42°C (where module operating temperatures reach 62°C).
- **Bifaciality & Albedo Capture:** The modules exhibit a bifaciality factor of $80 \pm 5\%$. The rear side captures diffuse horizontal irradiance and ground-reflected albedo. Given Purulia's terrain (dry gravelly soil with an albedo coefficient of ~0.20), rear-side irradiance adds a continuous 8% to 10% current boost (Imp increases from 16.36 A to ~17.80 A under peak conditions), elevating effective string output without increasing thermal voltage stress.

1.2 String Configuration & Voltage Safety Limits

The DC array consists of **4,340 total procured modules** (140 pallets of 31 modules each), deployed as **4,312 active modules** across **154 parallel strings of 28 modules in series**, with **28 cold spares** retained in the site depot.

```

[ 28 Modules in Series ]
(Voc = 51.20V STC -> 53.12V @ 10°C)
+----+ +----+ +----+ +----+
| M |->| M |->...->| M |->| M |----+
+----+ +----+ +----+ +----+ |
|
[ 154 Parallel Strings Total ] +---> [ Underground DC Bus ]
| (Max Voc: 1,487.36V DC)
+----+ +----+ +----+ +----+ | (Operating Vmp: 1,198V DC)
| M |->| M |->...->| M |->| M |----+
+----+ +----+ +----+ +----+

```

String Voltage Sizing Calculations

- **STC Open-Circuit Voltage (Voc at 25°C):** 51.20 V DC
- **Cold Winter Extreme Temperature (10°C in Purulia):** Temperature differential $\Delta T = 10^{\circ}\text{C} - 25^{\circ}\text{C} = -15^{\circ}\text{C}$.
- **Voltage Temperature Coefficient (β_{Voc}):** $-0.25\%/^{\circ}\text{C}$ ($-0.128\text{ V}/^{\circ}\text{C}$).
- **Maximum Cold Voc per Module:** $\text{Voc}(10^{\circ}\text{C}) = 51.20\text{ V} + (-15 \times -0.128\text{ V}) = 53.12\text{ V DC}$
- **Maximum Cold String Voltage at 10°C:** $\text{V}_{\text{string,max}} = 28\text{ modules} \times 53.12\text{ V} = \mathbf{1,487.36\text{ V DC}}$

Voltage Safety Verdict: This configuration enforces a strict safety margin below the **1,500 V DC** insulation ceiling of IEC 61730 Class A standards, DC solar cables, and inverter MPPT input limits, eliminating over-voltage insulation breakdown risks during cold winter morning startups.

1.3 Inverter Architecture & Closed-Loop Thermal Dynamics

The project utilizes the **FIMER PVS980-CS 4.3 MVA Compact Station**, housing a single high-power **PVS980-58 central inverter core**, a step-up transformer, and medium-voltage switchgear inside a 40-foot High Cube containerized steel enclosure.

```

+-----+
|                                     |
|               FIMER PVS980-CS COMPACT STATION               |
|                                     |
| +-----+ +-----+ +-----+ +-----+ |
| | PVS980-58 INVERTER | | MV TRANSFORMER | | SF6 SWITCHGEAR | |
| | - 1500V DC Input   | | - 600V / 33 kV   | | - 630A VCB     | |
| | - Thermosiphon Cool |---->| - ONAN Oil Immersed |---->| - REJ603 Relays |==> 33 kV Line
| | - 4,348 kVA @ 25°C | | - 3,000 kVA Continuous| | - 25 kA / 3 sec | |
| +-----+ +-----+ +-----+ +-----+ |
+-----+

```

Thermosiphon Closed-Loop Cooling Physics

Unlike traditional forced-air or liquid-pumped central inverters, the PVS980-58 employs a phase-change **thermosiphon cooling technology**:

- **Phase-Change Cycle:** A non-conductive dielectric liquid refrigerant absorbs heat directly from the insulated gate bipolar transistor (IGBT) power modules, vaporizing at low temperatures.
- **Passive Circulation:** The vapor rises naturally to an external air-cooled condenser heat exchanger mounted on the outer enclosure wall. Ambient air cools the condenser fins, causing the vapor to condense back into a liquid state and drain by gravity back to the IGBT cold plate.
- **Operational Reliability:** This self-contained phase-change loop operates without mechanical liquid pumps, valves, or active plumbing, eliminating coolant leak vulnerabilities and lowering auxiliary parasitic power consumption to $\leq 2,500$ W during full-load operation.

AC Output Thermal Derating Profile

AMBIENT TEMPERATURE	COMPACT STATION AC RATING	3.0 MVA GRID EXPORT MARGIN
At 25°C Ambient	4,348 kVA	+1,348 kVA Headroom (Zero Clipping)
At 35°C Ambient	4,229 kVA	+1,229 kVA Headroom (Zero Clipping)
At 50°C Peak Summer	3,845 kVA	+845 kVA Headroom (Zero Clipping)

1.4 Step-Up Transformer & 33 kV Switchgear Specifications

- **Transformer Type:** ONAN (Oil Natural Air Natural) mineral-oil-immersed outdoor transformer.
- **Voltage Ratio:** 0.615 kV low-voltage inverter output stepped up to 33 kV high-voltage grid export.
- **Vector Group & Winding:** Dyn11 vector configuration, with primary delta winding suppressing 3rd-order harmonic currents from entering the utility grid. Winding material is high-grade electrolytic copper/aluminum with an off-load tap changer ($\pm 2 \times 2.5\%$).
- **Medium Voltage Switchgear:** Compact SF6-insulated metal-enclosed switchgear panel housing a 630 A, 25 kA (for 3 seconds) vacuum circuit breaker (VCB).
- **Protection Relay Scheme:** Microprocessor-based ABB REJ603 protection relay providing instantaneous overcurrent (ANSI 50), time-delay phase overcurrent (ANSI 51), earth fault (ANSI 50N/51N), and transformer Buchholz/temperature trip outputs.

02 Civil, Structural & Meteorological Mechanics

2.1 Geotechnical Soil Analysis & Foundation Engineering

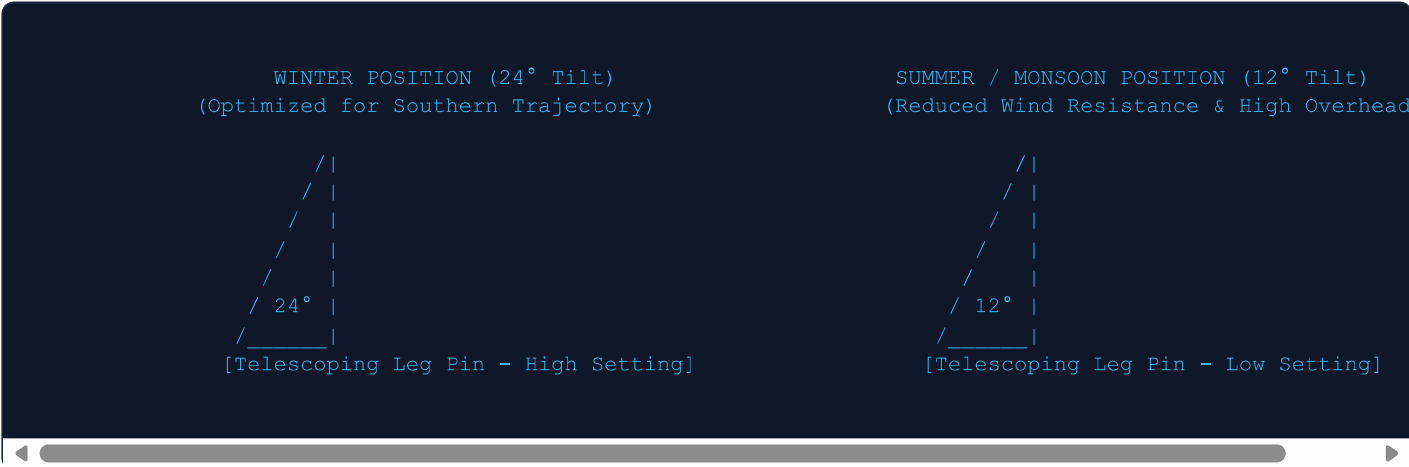
The project site in Santuri, Purulia district features typical Chota Nagpur plateau geology, characterized by hard gravelly red soil underlaid by weathered granite-gneiss rock at shallow depths (1.2 m to 2.0 m).

```
Surface Level (0.0 m)
===== <-- Grade Level
| Gravelly Red Soil / Sandy Clay |
| (Allowable Bearing Capacity: ~120-150 kN/m²) |
|----- <-- 1.0 m Depth
| Weathered Granite-Gneiss Bedrock |
| (High Passive Resistance & Lateral Stability) |
|----- <-- Pile Tip (1.5 m Depth)
| Reinforced Concrete Bored Pile (M30 Grade) |
| Diameter: 300 mm | Galvanized Steel Post Anchored |
=====
```

- **Pile Foundation Specification:** Reinforced concrete bored cast-in-situ piles using M30 grade concrete with sulphate-resistant cement.
- **Pile Dimensions:** Diameter 300 mm, driven to a minimum depth of 1.5 meters into the native soil/rock matrix.
- **Pull-Out & Uplift Resistance:** Designed to resist tensile uplift forces generated by high monsoon wind suction. Each pile foundation undergoes mandatory pull-out testing to withstand a minimum vertical uplift force of 18 kN and a lateral shear force of 12 kN.

2.2 Structural Load Dynamics & Seasonal Tilt Mechanics

The mounting structures utilize **Hot-Dip Galvanized Iron (HDG)** structural steel sections with a zinc coating thickness of ≥ 80 microns ($\sim 570 \text{ g/m}^2$), providing a corrosion resistance lifespan exceeding 25 years in atmospheric corrosion category C3. Certified via STAAD.Pro structural analysis to withstand basic wind speeds up to **170 km/h** (IS 875 Part 3: 2015).



Seasonal Yield Advantage: Shifting tables from 12° (Summer/Monsoon) to 24° (Winter) yields an additional **3% to 5% annual energy generation** (+135,000 kWh to +225,000 kWh per year) compared to a single fixed 23° installation, requiring only 3 days of low-cost manual labor twice a year.

2.3 Water Hydrology, Demineralization & Soiling Management

Dust accumulation (soiling) on glass surfaces reduces light transmittance, causing an unmanaged generation loss of up to 0.5% per day during dry months in Purulia.

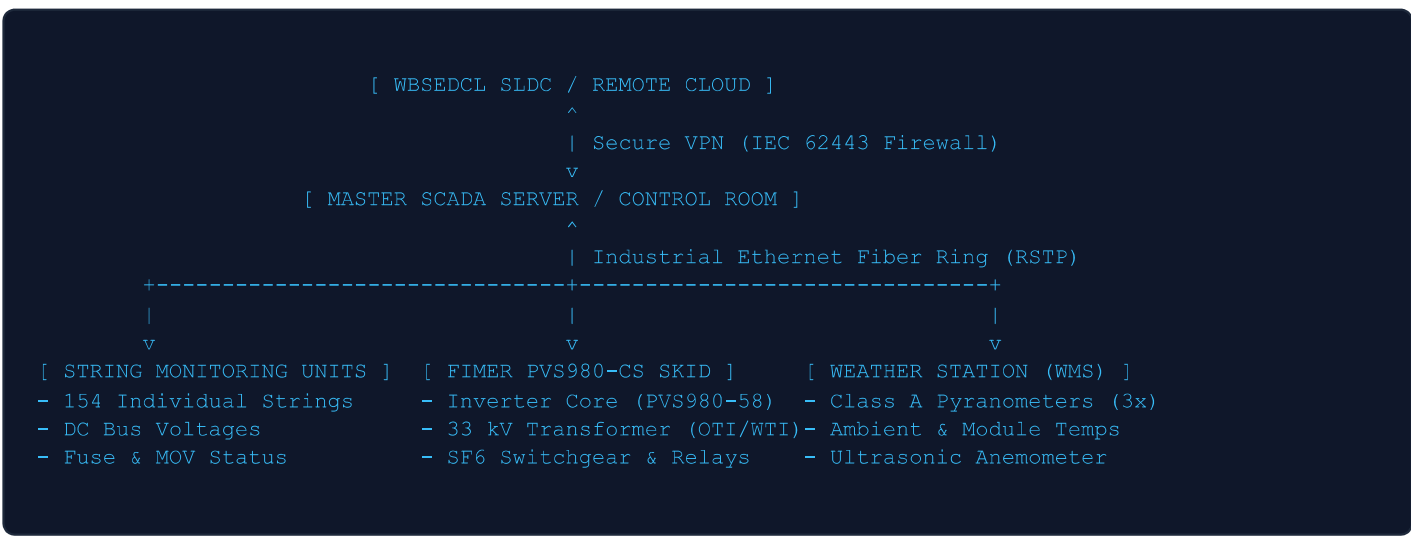
+-----+ Deep Borewell Raw Water Source Total Dissolved Solids > 450 ppm +-----+	====>+ Dual-Media Filter Suspended Solids Removal +-----+	====>+ Reverse Osmosis (RO) Desalination Unit TDS Reduction < 50 ppm Target +-----+	====>+ Demineralized Storage Tank TDS < 50 ppm pH 6.5 - 7.5 +-----+
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- **On-Site RO Plant:** On-site 150 mm deep borewell (5 HP pump, 5,000 L/hr) paired with a Dual-Media Filter and Reverse Osmosis plant to reduce TDS from ~450 ppm to **< 50 ppm** (pH 6.5–7.5), preventing calcification and permanent glass scaling.
- **Cleaning Schedule:** 24 cleaning cycles per year (bi-monthly during non-monsoon months). Each cycle consumes 2.5 liters of DM water per module (10,780 liters per full plant wash) at 20 bar pressure.

03 Monitoring, Control & Data Acquisition Infrastructure

3.1 SCADA System Architecture & Network Topology

The plant employs a **Master Supervisory Control and Data Acquisition (SCADA)** system built on industrial PLC/IPC hardware running IEC 60870-5-104 and Modbus TCP/IP protocols.



- **Redundant Fiber Backbone:** Single-Mode Fiber Optic (SMFO) ring running Rapid Spanning Tree Protocol (RSTP) ensures < 20 ms communication failover and complete EMI immunity across the 11-acre site.
- **Master Industrial PC:** Stationed in the site control room with RAID-1 SSD storage for 5 years of local logs and a 2-hour UPS battery backup.

3.2 String Monitoring Units (SMU) & Array Diagnostics

Data is collected at the **String Monitoring Unit (SMU)** level across all **154 parallel strings** using Hall Effect current sensors (±0.5% accuracy).

```
String 1 (28 Modules) ---> [ Hall Sensor 1 ] --+
String 2 (28 Modules) ---> [ Hall Sensor 2 ] --|
...
String N (28 Modules) ---> [ Hall Sensor N ] --+
                                |--> [ Local Microprocessor ] --> Modbus RS485 --> Fibe
                                    - Voltage Sensor
                                    - Temp Sensor
                                    - MOV Surge Monitor
```

- **Blown Fuse Alert:** Triggers instantly if string current drops to 0 A while parallel strings draw > 10 A.
- **Soiling / Shading Alarm:** Triggers if a string's output deviates by > 8% from neighboring strings under uniform irradiance.
- **PID / Degradation Tracking:** Tracks slow-onset current loss across strings over consecutive months.

3.3 Meteorological Station & Performance Accounting

SENSOR / METER	SPECIFICATION	PRIMARY FUNCTION
POA Pyranometer (Front)	Class A Thermopile (ISO 9060:2018), heated	Measures Plane of Array irradiance at 12°/24° tilt
POA Pyranometer (Rear)	Class A Thermopile, downward facing	Measures ground-reflected albedo gain
Module Temp Sensors	Pt100 4-wire RTD (4 units glued to modules)	Calculates cell temp and thermal voltage derating
Ultrasonic Anemometer	Solid-state ultrasonic sensor (no moving parts)	Measures wind speed (0 to 60 m/s) and direction
ABT Tariff Meters	Dual 0.2s Class Main/Check Tri-Vector Meters	Revenue accounting & GPRS AMR streaming to SLDC

$$PR = [E_AC_Exported (kWh) / P_DC_Installed (kWp)] \div [H_POA (kWh/m^2) / G_STC (1 kW/m^2)] \times 100\%$$

If the 15-minute rolling Performance Ratio (PR) drops below **75.0%** under unshaded conditions (POA > 600 W/m²), an automated SCADA high-priority alert is raised.

3.4 Site Security & Cybersecurity Architecture

- **Perimeter CCTV:** 12–16 HD IP Bullet Cameras with IR night vision (50 m) plus 2 High-Speed 360° PTZ Dome Cameras (30x optical zoom) with AI motion analytics.

- **IEC 62443 Cyber Security:** Industrial Stateful Inspection Firewalls segregate OT networks from IT/Internet traffic. Remote access uses encrypted IPSec/OpenVPN tunnels with MFA. SLDC telemetry uses outbound-only read-only channels.

04 Financial Engineering, Tax Code & Treasury Dynamics

4.1 Capital Expenditure Engineering Breakdown

The project requires a consolidated capital outlay of **₹10.80 Crore**, structured to achieve a debt-to-equity ratio of **70:30**.

CAPEX COST CENTER	GRANULAR COMPONENT SCOPE	TOTAL COST	% SHARE
PV Modules	4,340 Waaree 700W N-type TOPCon Modules (140 pallets)	₹5.60 Crore	51.9%
Power Conversion	FIMER PVS980-CS 4.3 MVA Compact Station & Switchgear	₹2.00 Crore	18.5%
Civil & Structural	Piling, HDG seasonal tilt tables, boundary wall, roads, store	₹1.10 Crore	10.2%
Grid Evacuation	3.0 km dedicated 33 kV line, poles, 33 kV bay at WBSEDCL	₹0.90 Crore	8.3%
Land Procurement	11.0 Acres contiguous freehold land & conversion fees	₹0.60 Crore	5.5%
Soft Costs & Fees	Engineering, CEIG inspection fees, insurance, contingency	₹0.60 Crore	5.6%
TOTAL CAPEX	COMPLETE TURNKEY PROJECT INFRASTRUCTURE	₹10.80 CRORE	100.0%

4.2 Capital Financing Structure

- **Promoter Equity Contribution (30%):** ₹3.23 Crore
- **Green Energy Bank Loan (70%):** ₹7.57 Crore at 8.50% p.a. over 7 years
- **Annual Bank Debt Service (EMI):** ₹1.58 Crore per year

4.3 Income Tax Code Optimization (Section 32 & Section 115BAA)

```
Gross Revenue (Yr 1): ₹1.83 Cr
|
v
(-) Plant OpEx: -₹0.10 Cr ==> EBITDA: ₹1.73 Cr
|
v
(-) Bank Interest (Sec 36): -₹0.62 Cr
|
v
(-) Accelerated Depreciation (Sec 32 - 40% WDV): -₹3.52 Cr
|
v
Taxable Corporate Income: -₹2.41 Cr (Net Operating Loss Carried Forward)
|
v
CORPORATE TAX PAYABLE (Years 1 to 6): ₹0.00
```

- **Accelerated Depreciation (Section 32):** Claiming 40% WDV accelerated depreciation in Year 1 (~₹3.52 Cr) creates an artificial tax shield, resulting in **₹0 corporate income tax liability for Years 1 through 6**.
- **Concessional Tax Rate (Section 115BAA):** Opting into Section 115BAA sets an effective flat corporate tax rate of **25.17%** (22% base + 10% surcharge + 4% cess) with complete Minimum Alternate Tax (MAT) exemption.

4.4 Corporate Treasury Wealth Compounding Model

Starting in Year 8 (post-debt clearance), the company deploys an unencumbered net operational cash surplus of **₹1.05 Crore per year** into a Corporate Treasury Mutual Fund portfolio for 17 consecutive years (Years 8 through 25) assuming a **12.0% CAGR**.

$$FV = P \times [((1 + r)^n - 1) / r] \times (1 + r) \implies FV = 1.05 \times [((1.12)^{17} - 1) / 0.12] \times 1.12 = ₹56.50 \text{ Cr to } ₹58.20 \text{ Cr}$$

- **Total Principal Invested:** 17 years × ₹1.05 Cr = ₹17.85 Crore
- **Pure Investment Capital Gain:** ₹38.65 Crore to ₹40.35 Crore

4.5 Terminal Company Liquidation Waterfall (Year 26)

```
[ TOTAL COMPANY GROSS ASSETS: ₹60.00 Cr - ₹61.70 Cr ]
- Mutual Fund Portfolio: ₹56.50 Cr - ₹58.20 Cr
- Scrap Steel, Copper & 11 Acres Land: ₹3.50 Cr
  |
  |--> (-) Corporate LTCG Tax on MFs (12.5% under Sec 112): -₹5.60 Cr to -₹5.80 Cr
  |--> (-) Corporate Tax on Asset Scrap (25.17%): -₹0.90 Cr
  |
  v
[ NET DISTRIBUTABLE COMPANY CASH BALANCE: ₹53.50 Cr - ₹55.10 Cr ]
  |
  +-----+
  |               |
  |               |
  v               v
[ OPTION A: Single-Year Exit ]      [ OPTION B: 5-Year Staggered Exit ]
- Dividend Tax (Sec 2(22)(e)): 39%    - Capital Reduction / Buyback Spread
- Tax Outflow: -₹20.70 Cr to -₹21.10 Cr - Tax Outflow: -₹16.40 Cr to -₹16.60 Cr
- NET IN-HAND: ~₹32.80 Cr - ₹34.00 Cr - NET IN-HAND: ~₹37.10 Cr - ₹38.50 Cr
```

05 Regulatory Code, Statutory Permitting & Grid Interconnection

5.1 WBLR Act Section 4C Land Conversion

Developing a 3.02 MWp plant on 11.0 acres in Santuri requires formal land conversion under **Section 4C of the West Bengal Land Reforms Act, 1955** from *Agricultural* to *Industrial - Solar Power Generation*.

```
[ Land Title Acquisition ] ==> [ Application to DL&LRO Purulia ] ==> [ BL&LRO Site Inspection ]
- 11.0 Acres Freehold          - Form 1 under Sec 4C          - Verify Non-Irrigated
- Deed Registration            - Mutation Certificates      - Confirm Single Crop
                                   |
                                   v
[ Conversion Clearance Issued ] <== [ Clearance from WB Land Board ] <== [ Payment of Conversion Fee ]
- Issued within 60 Days          DL&LRO Clearance          - Rs 15,000 / Acre
```

5.2 Grid Interconnection Standards & WBSEDCL 2025 Code

- **Harmonic Limits:** Central inverter maintains Total Current Harmonic Distortion (THDi) < 3.0%, complying with CEA Technical Connectivity Standards.
- **Anti-Islanding Protection:** Disconnects from the grid in < 100 ms upon utility outage to protect line maintenance crews.
- **Ride-Through Capability:** Features Low Voltage Ride Through (LVRT) down to 15% voltage for 150 ms and operates across 47.5 Hz to 52.0 Hz grid frequency limits.

5.3 Statutory Electrical Safety Mandate (CEIG Clearance)

Mandatory CEIG Inspection: Under Regulation 43 of CEA Safety Regulations 2010, installations above 11 kV cannot be self-certified. A physical site audit by the **Chief Electrical Inspector to the Government of West Bengal (CEIG)** is legally mandatory prior to energization.

Under WBSEDCL Procedures-C (2025), plants exceeding 170 kW capacity must undergo mandatory safety re-inspection and submit compliance certificates **every two years** throughout their 25-year operational lifecycle.

06 Consolidated Project Life-Cycle Comparison

PARAMETER	INITIAL BASELINE MODEL	FULLY OPTIMIZED UTILITY MODEL	ENGINEERING ADVANTAGE
DC Array Capacity	3.00 MWp (4,290 Modules)	3.02 MWp (4,340 Modules)	+18.4 kWp higher generation
String Sizing	143 Strings × 30 Modules	154 Strings × 28 Modules	Eliminates 1,500V cold-winter overvoltage
Power Conversion	2.5 MVA Inverter Skid	FIMER PVS980-CS 4.3 MVA Skid	3.845 MVA AC rating even at 50°C summer
Grid Evacuation	2.50 MVA Interconnection	3.00 MVA Dedicated Feeder	Zero AC thermal clipping headroom
Plant Monitoring	Basic Inverter Display	Fiber SCADA + SMU + Class A WMS	154-string diagnostic monitoring
Capital Expenditure	₹10.10 Crore	₹10.80 Crore	Includes full 3 MVA line & 4,340 modules
Financing Split	₹3.03 Cr Equity / ₹7.07 Cr Loan	₹3.23 Cr Equity / ₹7.57 Cr Loan	Maintains optimal 70:30 debt leverage
PPA Tariff Rate	₹3.80 / kWh Baseline	₹3.89 / kWh WBERC Rate	+₹4.00 Lakhs higher top-line per year
Year 25 Treasury Corpus	₹54.00 Crore	₹56.50 Cr to ₹58.20 Cr	+₹2.50 Cr higher mutual fund accumulation
YEAR 26 NET HARVEST	₹31.30 CRORE	₹32.80 CR TO ₹38.50 CR	UP TO +₹7.20 CR HIGHER NET LIQUID WEALTH